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Portfolio Website: nadeaujonny.github.io

Dear Hiring Manager,

I am writing to apply for your analyst opening. I am a UCSB mathematics graduate and Google Data Analytics certified, currently a Data Analyst Intern at Escape Communications, a defense and telecom electronics manufacturer in Torrance, where I have been the sole developer on two internal data tools. Both started as ambiguous requests and ended as working systems in the hands of the people who asked for them, which is the clearest evidence I have of how I work.

The first was a failure analysis search tool. Escape files a record for every hardware failure that comes back, and roughly 4,038 had accumulated in a legacy Bugzilla instance from 2007 through 2026, configured years earlier by an administrator who left without documentation. Nobody remaining knew how to get data out of it. I reverse-engineered the data model and, after direct database access was declined, built the extraction against the REST API rather than treating that as a workaround. The most consequential thing I did was an audit nobody asked for: the original plan was a dashboard over the structured failure-classification fields, and I found those fields could not support one. Forty-three percent of records were structurally unable to carry a failure-type value at all, and a second field defaulted to a single value in over eighty percent of cases. The real engineering knowledge was in the free-text comment threads, so I redirected an eight-week roadmap toward search rather than build a dashboard over noise. From there I wrote a dependency-free Python pipeline that indexes roughly 2,245 PCB reference designators out of that free text, hand-validated it prefix by prefix, and published a per-category precision table rather than a single headline accuracy number. That validation surfaced collision classes I would never have predicted and produced a fix that eliminated a 68 to 71 percent false-positive rate on the worst-performing values.

The second was a BOM sourcing and purchase planning tool. Buying parts for a board build meant hand-cleaning a 300-line spreadsheet, uploading part lists to each distributor's site one at a time, and comparing quotes by eye, with no record of why anything was bought. I built a Python tool with 444 automated tests that cleans and validates the bill of materials, checks availability and pricing across five distributors through file parsers and live REST APIs, and produces a cost-minimized purchase plan with a full audit trail. Wiring in live vendor APIs exposed volume price breaks the old file-based process structurally could not see, cutting a 100-board plan from \$129,770 to \$103,774. A parametric specification validation layer, comparing each row's stated electrical characteristics against the actual manufacturer part number, found four genuine documentation defects in production BOM data on its first run, and availability checks flagged 55 at-risk parts ahead of the build rather than during it. When a cost comparison later looked suspiciously favorable, I traced it and found the single-vendor baseline was silently inheriting costs from lines that vendor could not actually supply. The arithmetic was correct and the label was not, so I fixed the presentation, added coverage statistics, and changed no numbers.

Both tools also shipped with the documentation to outlive me: user guides written for non-technical readers, runbooks, and append-only decision logs recording why each choice was made. That was deliberate, since the first project existed precisely because a prior administrator left without any. I briefed the VP of Engineering on the sourcing tool in a one-page readout that led with the cost comparison rather than technical progress, and closed with the integrations I had assessed and recommended against, reasoning attached.

Behind that work is a mathematics degree from UCSB, the Google Data Analytics Professional Certificate completed with a 100 percent average across all nine courses, and UCLA Extension's Data Science Certificate in progress through December. I also spent 1,500-plus hours teaching mathematics, which is where I learned to give the same finding at three different depths depending on who is asking. My portfolio of fifteen projects spanning SQL, Excel modeling, Power BI, Tableau, Python, and R is at nadeaujonny.github.io, including an Oracle SQL and Power BI supply chain build on 180,000 orders and a three-statement FP&A model built from 10-K filings.

Thank you for your time and consideration. I would welcome the chance to talk about the role.

Sincerely,

Jonathan Nadeau